

NIKHIL KHATRI

Software Developer | Java Backend | AI Integration | Spring Boot | Microservices

+91 8104158111 | nikhilkhatri017@gmail.com | linkedin.com/in/nikhilkhatri017 | github.com/nikhilkhatri017 | [LeetCode](#)

PROFESSIONAL SUMMARY

Motivated Computer Science graduate with hands-on experience building production-ready Java and Spring Boot microservices, RESTful APIs, and AI-integrated backend systems. Proficient in designing scalable architectures using Spring Cloud, Keycloak OAuth2, RabbitMQ, Docker, and multi-database strategies (PostgreSQL + MongoDB). Demonstrated ability to integrate AI/ML models (Google Gemini API, LangGraph, Groq LLMs, TensorFlow) into real-world applications. Strong foundation in Data Structures, Algorithms, OOP, and Software Engineering principles. Eager to contribute as a backend developer, Java developer, or AI integration engineer in a fast-paced environment.

TECHNICAL SKILLS

Languages: Java, Python, SQL

Frameworks & Libraries: Spring Boot, Spring Cloud, Spring MVC, Spring Security, Microservices, RESTful API, Hibernate/JPA, FastAPI, LangChain, LangGraph, React, Redux

AI / ML: TensorFlow, Keras, Scikit-learn, OpenCV, Google Gemini API, Groq LLMs, Hugging Face, Pandas, NumPy, Matplotlib, CNN, NLP, LLM Integration

Messaging & Auth: RabbitMQ, Keycloak OAuth2, JWT

Databases: PostgreSQL, MongoDB, MySQL, SQL

DevOps & Cloud: Docker, Kubernetes (basic), AWS (basic), Maven, Git, GitHub, GitHub Actions (CI/CD)

Tools & IDEs: IntelliJ IDEA, VS Code, Jupyter Notebook, Anaconda, Postman

CS Fundamentals: Data Structures & Algorithms, OOP, OS, DBMS, Computer Networks, Software Engineering

PROJECTS

Fitness Microservices Platform

2024 – Present

Java | Spring Boot | Spring Cloud | Keycloak | RabbitMQ | PostgreSQL | MongoDB | Google Gemini API | Docker | React | Redux | MUI

GitHub: github.com/nikhilkhatri017/fitness-microservice

- Designed and implemented a full-stack fitness application using microservices architecture with 6 independent Spring Boot services: Eureka Service Registry, Config Server, API Gateway, User Service, Activity Service, and AI Service.
- Integrated Google Gemini API (2.5 Flash) to deliver AI-powered personalized fitness recommendations via a dedicated microservice, demonstrating end-to-end AI integration in a Java backend.
- Implemented centralized authentication and authorization using Keycloak OAuth2 and configured RabbitMQ for asynchronous, event-driven inter-service communication.
- Applied a polyglot persistence strategy: PostgreSQL for relational user data and MongoDB for flexible document storage of activity logs and AI recommendations.
- Built a React + Vite frontend with Redux Toolkit for state management and Material UI (MUI) components; configured CORS support in the API Gateway for seamless frontend-backend communication.
- Java codebase exceeds 44,000+ lines; repository actively maintained with Docker containerization support and roadmap for Kubernetes orchestration and CI/CD pipelines.

AI-First CRM HCP Module

2025

Python | FastAPI | LangGraph | LangChain | Groq LLMs | SQLAlchemy | PostgreSQL | React | Redux | Vite

GitHub: github.com/nikhilkhatri017/ai-first-crm-hcp

- Built a conversational AI-powered CRM system for healthcare professionals that eliminates manual form entry by using Large Language Models (Groq: gemma2-9b-it, llama-3.3-70b-versatile) to extract structured interaction data from natural language input.
- Architected a multi-node LangGraph agentic workflow with tools for intent classification, interaction logging, form editing, compliance validation, follow-up suggestion, and interaction summarization.
- Developed RESTful API backend using FastAPI with SQLAlchemy ORM and PostgreSQL; implemented CORS middleware for React frontend integration.
- Frontend built with React 18, Redux Toolkit, and Vite; features a split-panel UI where the AI chat automatically populates the interaction form in real time.

- Incorporated built-in pharmaceutical compliance checks to flag risky or off-label language in sales representative interactions.

JournalApp – Secure E2EE Spring Boot Application

2024 – Present

Java | Spring Boot | Spring Security | MongoDB | Maven | GitHub Actions CI/CD

GitHub: github.com/nikhilkhatri017/journalApp

- Developed a secure end-to-end encrypted (E2EE) personal journal web application using Java and Spring Boot, with MongoDB as the backend data store.
- Implemented Spring Security for authentication/authorization; configured GitHub Actions CI/CD workflow for automated build and deployment pipeline.
- Applied RESTful API design principles for CRUD operations on journal entries with proper security constraints.

Hand Sign Language Recognition System

2023 – 2024

Python | TensorFlow | Keras | CNN | OpenCV | Jupyter Notebook

- Developed a real-time hand sign language recognition system using Convolutional Neural Networks (CNN) with a hand tracking preprocessing pipeline built on OpenCV.
- Implemented binary image preprocessing to isolate hand boundaries and trained classification models to map hand sign symbols to their corresponding meanings.
- Explored model ensemble techniques combining multiple CNN models on the same input to improve classification accuracy; optimized the model for resource-constrained hardware.

Thyroid Disease Detection System

2023

Python | Scikit-learn | TensorFlow | Pandas | NumPy | Matplotlib | XGBoost | Jupyter Notebook

- Implemented and benchmarked 10+ ML models including Logistic Regression, KNN, SVM, Decision Tree, Random Forest, Naive Bayes, DBSCAN, XGBoost, and ANN for thyroid disease detection.
- Performed advanced data preprocessing, feature selection, and model evaluation; selected the best-performing model based on F1-Score and Accuracy metrics.

Rock Paper Scissors – Java Backend Game

2024

Java | Spring Boot | RESTful API

GitHub: github.com/nikhilkhatri017/RockPaperScissor_JavaBackend

- Built a RESTful Java Spring Boot backend service implementing game logic for Rock Paper Scissors with clean API endpoints and stateless design principles.

EDUCATION

Bachelor of Technology – Computer Science and Engineering

2021 – 2025

Silicon Institute of Technology, Silicon University | Bhubaneswar, Odisha

CGPA: 7.97 / 10

Intermediate (Class XII)

2020 – 2021

Sankalp Academy | Jhalawar, Rajasthan

73%

Matriculate (Class X)

2017 – 2018

Lady Anusuiya Singhan Education Center | Jhalawar, Rajasthan

80%

CERTIFICATIONS & COURSES

- Deep Learning – IIT Ropar, NPTEL (2025)
- Generative AI – Syllogistek Systems Private Limited (2024)
- Data Science, Machine Learning & Deep Learning using Python – Syllogistek Systems Private Limited (2023)
- Full Stack PHP – Syllogistek Systems Private Limited (2022)

POSITIONS OF RESPONSIBILITY

Joint Secretary – University Student Council

2023 – 2024

Silicon University, Bhubaneswar

Awarded Best Council Member 2024

- Led inter-departmental coordination, managed student affairs, and represented student interests at institutional level.

Member – Green Club

- Organized multiple campus tree plantation drives to promote sustainability and environmental awareness.

Member – Blood Donation Club

- Participated in blood donation campaigns and awareness drives on campus.